Ali Nawaz

Senior AI/ML Engineer

Islamabad, Pakistan | Open to Remote | Open to Full-Time | Freelance | Research | Advisory

GenAI | Agentic AI | RAG | LLM | Computer Vision | NLP | Python | ML

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PROFESSIONAL SUMMARY

Senior AI/ML Engineer with 6+ years building and deploying production AI systems at scale across healthcare, government, and enterprise. Specializes in agentic AI and multi-agent systems, LLM and RAG applications, and computer-vision pipelines, with strong command of LangChain, LangGraph, Azure (OpenAI, AI Foundry, AI Search), GCP, AWS, and MLOps. Has shipped LLM-powered automation platforms, medical-imaging CAD systems trained on clinical datasets, and a national-scale government workflow system for FIA operations, all built to be reliable, explainable, and maintainable in production. Research background spans clinical and ophthalmic machine learning across major medical-imaging modalities, including X-ray, MRI, and CT. Recent research output: one paper accepted at IFIP IoT 2026, a first-author journal paper under review at IEEE JBHI, two papers under review at ICCII 2026 (one first-author), and a mammography vision-language manuscript in preparation for an IEEE journal. Open to Senior, Lead, or Principal AI Engineer roles where deep technical expertise meets high-impact product delivery.

CORE SKILLS

- **Languages:** Python, C++, C, TypeScript, SQL
- **AI / ML:** Agentic AI, Multi-Agent Systems, LLMs, RAG, Computer Vision, NLP, GenAI, MLOps
- **ML Frameworks:** PyTorch, TensorFlow, Scikit-learn, LightGBM, Hugging Face
- **LLM & Agents:** LangGraph, LangChain, Azure AI Foundry, Azure AI Search, Azure OpenAI, ChromaDB, RAG pipelines
- **Backend:** FastAPI, Flask, NestJS, REST APIs, RBAC, WebSockets
- **Frontend:** Next.js, React, TypeScript, HTML5, CSS, Tailwind CSS
- **Databases:** PostgreSQL, Prisma, SQLite, Redis, Elasticsearch
- **Cloud & DevOps:** AWS EC2, GCP VM, Azure, NVIDIA GPU / CUDA / TAO Toolkit, Docker, GitHub Actions, GitLab CI, Azure DevOps
- **Computer Vision:** YOLOv8–v11, OpenCV, Imagenet Models, Object Trackers, Segmentation Networks, ST-GCN, PoseClassificationNet, C3D, PaddleOCR, Faster R-CNN
- **Real-Time / Streaming:** WebSocket, UDP, TCP, FFmpeg, h264
- **Project Delivery:** System Design, Agile / Scrum (Jira, ClickUp, Trello), Technical Documentation, Team Leadership

PROFESSIONAL EXPERIENCE (6+ Years)

PAK-AI, Islamabad, Pakistan (Remote) June 2025 – Present

CEO & Cofounder

Founded Pak-AI, an AI-first platform company building production SaaS, agentic AI systems, and research tools, leading a cross-functional team across AI, full-stack, and product.

Projects:

- **Pak-AI Platform - SaaS Website [Link]** | *Next.js, NestJS, PostgreSQL, Prisma, Docker*
  - Built **monorepo SaaS platform** across **5 phases**.
  - Developed **SEO-driven** content engine with **editorial workflows** and AI assistant integration.
  - Delivered **admin dashboard** for publishing, lead tracking, and **analytics**.

- **AZ Business Consultant [Link]** | *Next.js, TypeScript, Supabase, Prisma, Vercel*
  - Built a full-stack lead-gen website with an 8-line services mega-menu and ~65 dynamic sub-service pages.
  - Developed a lead-capture engine (multi-step wizard, WhatsApp, email) secured with auth, row-level security, and rate limiting.
  - Delivered an admin CMS for publishing, lead tracking, and content management across 10+ modules.

- **Alvina - Production Multi-Agent AI Platform** | *LangGraph, Azure OpenAI, FastAPI, NestJS, Next.js 15, PostgreSQL*

- Built production multi-agent platform with on demand **AI agents** creation for the specified task using **LangGraph (**Supervisor architecture**)**.
- Designed **full-stack system** web, desktop version powered by **GPT-5.4 (**Azure OpenAI & Foundry).
- Developed a **RAG pipeline** for citation- and reference-based answers using **ChromaDB** for semantic search and retrieval.
- Integrated with **agent memory** & **tool-calling.**
- Implemented **PostgreSQL + Redis** for scalable storage & session management.

- **FIA Smart ACR/PER Management System [Link]**| *NestJS, Next.js, PostgreSQL, Prisma, Docker, RBAC*
  - Led development of national-scale workflow platform digitizing **FIA ACR/PER** lifecycle.
  - Built state-machine workflow engine from draft → approval → archive with **hash-verified** records.
  - Implemented **role-based** workflows (RBAC) across actor types with auto-routing & notifications**.**
  - Designed multi-level org hierarchy with **scoped access** control & user lifecycle management.

Invozone, Johar Town Lahore, Pakistan (Remote) April 2025 – Present

Senior AI/ML Engineer

Projects:

- **R34L – AI-Powered Autonomous IT Healing Platform** [Link] | *Azure OpenAI/Search, LangGraph, Next.js, PostgreSQL, WebSockets*
  - Built **agentic AI automation engine** using LangGraph with reasoning loops, tool-calling, and safe execution of remediation workflows on Windows endpoints
  - Developed **Service Guard** safety layer enforcing policy gates, rollback logic, and crash-loop detection for production reliability
  - Implemented **adaptive learning model** using technician feedback, rollbacks, and success rates to refine decision pathways
  - Engineered secure **WebSocket system** for telemetry, orchestration, and human-in-the-loop approvals via Next.js dashboard.
  - Deployed and orchestrated the platform's LLM components on **Azure AI Foundry,** model deployment, integration, and lifecycle management.
  - Built the **RAG** retrieval layer with **Azure AI Search**, using vector and hybrid keyword + semantic search.
- **Mynatek – Hospital Cleaning Verification System** [Link] | *PyTorch, YOLOv11, ST-GCN*

  Led the AI development of an edge-based hospital cleaning verification system combining pose estimation, object segmentation, and action recognition for real-time hygiene monitoring.
  - **ST-GCN-based** action recognition system for classifying **cleaning vs non-cleaning activities** using **17-keypoint** skeletal motion sequences.
  - Built **YOLOv11 pose + ByteTrack** pipeline for **multi-person** tracking and consistent identity assignment across frames.
  - Designed **two-stream** model combining **pose + motion features** for improved temporal understanding.
  - Implemented **activity classification** (cleaning vs. non-cleaning) from **17-keypoint skeletal sequences** using NVIDIA **PoseClassificationNet (PCN)**, configuring the **NVIDIA TAO/CUDA GPU environment** for fine-tuning and inference.

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doAZ, Gangnam-gu, Seoul, South Korea (Project Based Contract - Remote) Oct 2024 – March 2025

Senior Computer Vision Engineer

- Led **data labeling** using **Roboflow** for detection and segmentation to refine AI model accuracy.
- Developed a customized **Template Matching** based **Annotator** for creating the Blueprint floor dataset.
- Endpoints for automatic symbols detection model for civil drawings using **Template Matching** and **YOLOv8**.
- Created the production-ready **Sobang 2.0** and **Sobang 3.0** application for the South Korean Government, enabling automatic symbol counting in civil engineering drawings for building approvals [Link].
- Contributed to the development of **PointChecker**, an AI-powered document analysis tool for automated review, citation, and structured extraction using **LLMs** and **RAG pipelines** [Link].
- Built **PDF parser** with different SOTA parser like Llamaparse, OpenAI, Gemini Flash 2.0, Marker, Docling, markitdown and Cohere Rerank 3.5 for accurate document processing
- Researched **GAN-based** image background removal and **Real-ESRGAN-based image restoration** to enhance blueprint clarity.

Horizon Tech Services, (NSTP, NUST) Islamabad, Pakistan July 2022 – Sep 2024
Lead AI Engineer ***(Dec 2023 – Sep 2024)*** | *AI Engineer* ***(Jul 2022 – Dec 2023)***

Led end-to-end development of multiple AI systems across media intelligence, surveillance, and applied research domains, delivering production-grade solutions using modern deep learning and NLP stacks.

Projects:

- **AI-Based Media Content Analysis System** *(Python, Django, PostgreSQL, PyTorch, Hugging Face, FFmpeg)*
  - Led the development of a comprehensive AI based Media Content Analysis system, managing electronic, print, and social media data:
  - **Electronic Media**: Directed **low-latency live streaming** for 64+ channels (Urdu/English) with 30 days backup and built real-time **video clipping**, Screenshots and tickers capturing with **OCR** capabilities.
  - **Print Media**: Automated news scraping from 50+ sources and improved data handling with Elasticsearch and e-newspaper analysis tools.
  - **Social Media**: Built **Twitter** and **YouTube** data extraction bots with NLP pipelines for **summarization,** sentiment analysis, and classification.
  - Implemented **ML pipelines** for multilingual transcription, translation**, OCR,** and text classification.
- **AI-Based Surveillance & Activity Recognition System** *(YOLOv8, YOLOv10, C3D, Custom CNN, OpenCV)*
  - Built **real-time surveillance AI system** integrating ANPR, face recognition, person detection, and activity recognition
  - Designed **C3D-based spatio-temporal** feature extraction pipeline for video understanding.
  - Developed **custom CNN models** for real-time activity **classification** from live camera feeds.
  - Enabled **multi-camera anomaly detection system** for public safety monitoring.
  - Supported AI projects with data labeling, model development, and automated annotation using advanced tools, improving team productivity.
- **Applied AI Research & Engineering Contributions**
  - Conducted **R&D in NLP and deep learning** for sentiment analysis, translation, and news classification
  - Improved YOLOv8/YOLOv10 object detection systems for real-time surveillance applications
  - Contributed to **aerial imagery projects (Uqab)** and smart surveillance model training pipelines
  - Implemented **image forgery** detection systems using **SIFT, SURF, IHC, AKAZE** methods
  - Developed adversarial robustness systems using **Defense-GAN** and generative models

Xylexa, (NSTP, NUST) Islamabad, Pakistan Dec 2020 – June 2022
AI Engineer

- **Medical Imaging based CAD Development:**
  - Built breast cancer detection by analyzing **mammograms** for **lesion identification** [link].
  - Developed **chest X-ray triage** and multi-disease classification models, including **lung segmentation** for **pneumonia detection** [link].
  - Utilized **multi-modality** datasets like **MIMIC-CXR,** VinDr-Mammo, VinDr-PCXR, and VinDr-SpineXR to enhance model accuracy.
  - Collaborated with radiographers for **annotation** and problem understanding, improving diagnostic precision.
  - Enhanced image quality and updated **Nifti** and **DICOM** tags.
  - **Integrated incremental** learning processes into existing pipelines for training and **finetuning** models.
  - CICD pipelines using **GitHub actions** for project version control and **Docker** for containerization.
  - Completed medical imaging certifications, including the **CITI** Program for Human Research.

Independent & Freelance Projects

- **Alvina - Production Multi-Agent AI Platform** | *LangGraph, Azure OpenAI, FastAPI, NestJS, Next.js 15, PostgreSQL*
  - Built production multi-agent platform with on demand **AI agents** creation for the specified task using **LangGraph (**Supervisor architecture**)**.
  - Designed **full-stack system** powered by **GPT-5 (**Azure OpenAI & Foundry).
  - Developed **RAG pipeline** with **ChromaDB** for semantic search & retrieval with referencing.
  - Integrated with **agent memory** & **tool-calling.**
  - Implemented **PostgreSQL + Redis** for scalable storage & session management.

- **Ilmevo AI-Driven Content Platform** | *Next.js, NestJS, PostgreSQL, Prisma, Docker*
  - Built a monorepo **SaaS** platform with an **SEO-driven** content engine, editorial workflows, integrated **AI assistant**, and an admin dashboard for **publishing**, lead tracking, and analytics.

University of Engineering and Technology, Taxila, Pakistan March 2019 – July 2020
Research Assistant (RA/TA)

- Started AI research at **SIMPLE Lab** during my master's degree, focusing on **brain MRI** analysis.
- Developed AI frameworks for medical imaging, with a focus on **Alzheimer's Disease** detection.
- Specialized in **2D** and **3D** imaging techniques, applying AI to improve diagnostic accuracy.

INTERNSHIPS (1.5 Years)

DYNIMAX Intermedia, Islamabad, Pakistan Jan 2017 – June 2017
Internship: KP IT Board Internship Program

- Completed **KP IT Board Internship Program**, gaining hands-on industry exposure.
- Master's in graphic design tools: **Photoshop**, **Illustrator**, **Corel Draw**, Sony Vegas, and **Canva**.
- Enhanced skills in graphic design, video editing, photography, and 2D/3D animation.

Cambridge Model College, Karak KPK, Pakistan March 2017 – Mar 2018
KP IT Board Internship Program
Internship: National Internship Program (NIP)

- Participated in the **National Internship Program** at CMC Karak.
- Collaborated with **KPK IT Board Peshawar**, contributing to software industry projects.
- **Senior ICT Instructor** teaching computer science and overseeing labs at Cambridge Model College, Karak

AI RESEARCH PROJECTS

- **Myopia Progression Prediction — Clinical Pipeline** *LightGBM, SHAP, SMOTE, Streamlit, Scikit-learn*
  - End-to-end clinical ML pipeline for predicting **myopia progression** from **corneal topography data.**
  - Conducted **feature engineering, preprocessing, SMOTE balancing**, and **11-model comparative study** with stacking ensemble.
  - Achieved **AUC-ROC 0.9960** using **LightGBM**, demonstrating clinically strong predictive performance
  - Integrated **SHAP-based explainability** for **patient-level risk** interpretation and clinical transparency.
- **Corneal Ectasia Classification Pipeline** *ResNet50, EfficientNet-B0, DenseNet121, PyTorch*
  - Developed deep learning classification system for detecting Keratoconus vs Normal corneas from tomography images.
  - Trained and compared multiple CNN architectures including **ResNet50, EfficientNet-B0, and DenseNet121** with Gradcam overlay on affected area.
- **IVCM Dry-Eye Research Platform** *SNP-Net, FastAPI, Segmentation, Morphometric Analysis*
  - Designed corneal IVCM research pipeline for dry-eye disease analysis and biomarker extraction
  - Implemented SNP-Net-based segmentation and morphometric analysis workflows
- **Bone Cancer Classification Whole-Body Bone Scans** *Data fusion, Dark-Net, ConvNeXt-Tiny + Swin-T encoders*
  - Built a deep learning model to classify bone scans as Normal, Metastatic, or Degenerative using paired Anterior + Posterior views on a 478-patient cohort.
  - Designed a cross-view attention fusion architecture over ConvNeXt-Tiny + Swin-T encoders to combine both views into a single diagnosis.
  - Delivered a production-grade PyTorch pipeline (DICOM ingestion, training, Grad-CAM explainability, Streamlit demo) following TRIPOD-AI standards.

EDUCATION

University of Engineering and Technology – Taxila, Pakistan 2017 – 2020
MS - Software Engineering.
CGPA - 3.21/4.00
Thesis Title: Deep Convolutional Neural Network-based Classification of Alzheimer's disease using MRI Data

Major Subjects: Advanced Machine Learning, Algorithm Design, Research Methodology, Software Architecture.

University of Science and Technology – Bannu, Pakistan 2012 – 2016

B.Sc. - Software Engineering.
CGPA - 3.53/4.00
FYP Title: Online Job Portal

Major Subjects: Artificial Intelligence, Data Structures, Software Engineering, Database Systems

AWARDS AND ACHIEVEMENTS

- **Prime Minister Laptop Scheme Award**: Recognized for ranking in the top 10% in Software Engineering at the University of Science and Technology Bannu.
- **Diya Foundation Scholarships**: Awarded for academic merit and financial need during bachelor's studies.
- **Best Student Award**: Honored for excellence in academic and extracurricular activities.
- **Fauji Foundation Scholarship**: Granted to beneficiaries for outstanding academic performance.

LEADERSHIP AND VOLUNTEERING

- **AI Seminars & Workshops:** Spearheaded sessions at different universities on AI, Machine Learning, and Data Science, engaging students and professionals.
- **Online Tutoring:** Volunteered to mentor underprivileged students in tech education and career development.
- **Community Leadership:** Directed a community of 550+ students, organizing workshops and coding sessions for skill enhancement.
- Collaborated on transformative projects "**Opti Key**" for blind people to utilize technology, and "**Bacteria Cell Counter**" heralding advancements in assistive technology for biotechnology field.

PUBLICATIONS

Journals:	Kim, Youngtae et al., "**Dual-Phased Detection and Classification of Auxiliary Insulation Using YOLOv8 Models**," Inscanner, Doaz Corporation & Lotte E&C, Seoul, South Korea, 2025. (https://arxiv.org/pdf/2502.18871)
	Manzoor, Sohail et al., "**Melanoma Detection Using a Deep Learning Approach**," International Journal of Innovations in Science & Technology, 2022.
Conference:	Nawaz, Ali et al., "**Deep Convolutional Neural Network-Based Classification of Alzheimer's Disease Using MRI Data,**" IEEE INMIC, 2020.
	Khan, et al., "**Optimized Prediction Model for Type 2 Diabetes Mellitus Using Gradient Boosting Algorithm,**" MAJICC, 2021.
	Iqbal, Javid et al., "**Ordinary Learning Method for Heart Disease Detection Using Clinical Data**," iCoMET, 2020.
	Ahmad, Fawad et al., "**Mortality Prediction in ICU Patients Using Machine Learning Models**," IBCAST, 2021.
	Rehman, Abdur et al., "**Benchmarking Temporal Multi-Channel CNN-LSTM Models for 24-Hour M-Class Solar Flare Forecasting Using SDOBenchmark**," IFIP IoT 2026 (accepted; to be presented).
Under Review:	Nawaz, Ali et al., "**Refining Saliency Maps through Region Proposals for Improved Weakly-Supervised Thoracic Disease Localization and Classification**," IEEE Journal of Biomedical and Health Informatics (under review).
	Nawaz, Ali et al., "**When Does Astrometric Uncertainty Help? Evaluating Monte Carlo HDBSCAN for Open Cluster Membership in Gaia DR3**," submitted to ICCII 2026.
	Javaid, Sameena et al., "**Which Lesions Need Dermoscopy? Value-of-Information Routing for Budgeted Second-View Skin Lesion Classification**," submitted to ICCII 2026.
In Preparation:	Javaid, Sameena et al., "**Conversational Mammography AI Under Audit: Label Priors, Dialogue Shortcuts, and the Value of the Image in CBIS-DDSM Question Answering**," to

be submitted to IEEE Transactions, 2026.

REFERENCES

Academic: Assoc. Prof. Engr. **Dr. Syed M. Anwar**, Medical Image & Signal Processing, University of Engineering and Technology, Taxila (UET), Pakistan. Email: s.anwar@uettaxila.edu.pk Contact: +92-300-5044036, Faculty Profile Page: [Link]

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